

Assignment 1 - Electricity Bill Calculator

Student Details

Student Name	Om Chavhan
Class	CS-N
Division	N
Roll Number	4
Assignment Name	Electricity Bill Calculator with Database

GitHub URL

<https://github.com/omudiii/WTassignment.git>

Project Description

A responsive, database-enabled Electricity Bill Calculator web application built with PHP, CSS, and SQLite. The system calculates electricity bills based on consumption units using progressive slab rates and stores all billing records in a database for historical tracking.

Technology Stack

- Backend: PHP 7.4+
- Database: SQLite
- Frontend: HTML5, CSS3, JavaScript
- Server: PHP Built-in Development Server

Features

1. Customer Details Collection
 - - Name, Consumer ID, Area/Locality, City
2. Bill Calculation with Progressive Slabs
 - - First 50 units: Rs. 3.50/unit
 - - Next 100 units: Rs. 4.00/unit
 - - Next 100 units: Rs. 5.20/unit
 - - Above 250 units: Rs. 6.50/unit
3. Database Storage
 - - All bills saved with complete customer and breakdown details
 - - Persistent storage across sessions

4. 4. Billing History & Summary

- - View all monthly bills with usage and amounts
- - Monthly summary with total units, total bill, average
- - Expandable slab breakdown for each bill

5. 5. Real-time Preview

- - Live bill calculation as you type

6. 6. Responsive Design

- - Works seamlessly on desktop, tablet, and mobile

Database Schema

SQLite Tables:

- bills table: Stores customer and bill information
- bill_breakdown table: Stores slab-wise breakdown for each bill

File Structure

- index.php - Main application with form and database integration
- db.php - Database connection, initialization, and helper functions
- style.css - Responsive styling for all components
- bills.db - SQLite database file (auto-created on first run)

Usage Instructions

1. Navigate to the project directory
2. Run: `php -S localhost:8000 -t .`
3. Open `http://localhost:8000/index.php` in browser
4. Enter customer details and units consumed
5. Click "Calculate" to generate bill and save to database
6. View billing history and monthly summary on same page

Output & Results

Once a bill is calculated:

- Displays customer details and billing breakdown
- Shows total units and total bill amount
- Saves record to database automatically
- Updates monthly summary table with aggregated data

- Adds entry to billing history with expandable slab details

Output:

Electricity Bill Calculator

Responsive, clear slab breakdown and instant preview

Customer Details

Name

Om Chavhan

Consumer ID

78963215

Area / Locality

Renuka nagar

City

Akola

Billing

Units Consumed

520

Calculate

Reset

Live Preview:

Estimated bill: Rs. 2850.00

Breakdown:
First 50 units: 50.00 units x Rs.3.50 = Rs.175.00
Next 100 units: 100.00 units x Rs.4.00 = Rs.400.00
Next 100 units: 100.00 units x Rs.5.20 = Rs.520.00
Above 250 units: 270.00 units x Rs.6.50 = Rs.1755.00

Result

Name: Om Chavhan

Consumer ID: 78963215

Location: Renuka nagar Akola

Slab	Units	Rate	Amount (Rs.)
First 50 units	50.00	3.50	175.00
Next 100 units	100.00	4.00	400.00
Next 100 units	100.00	5.20	520.00
Above 250 units	270.00	6.50	1,755.00

Total Units: 520.00

Total Bill: Rs. 2,850.00

✓ Bill saved to database!

Monthly Summary

Month	Records	Total Units	Total Amount (Rs.)	Avg Bill (Rs.)
August 2026	1	520.00	Rs. 2,850.00	Rs. 2,850.00

Billing History - All Months & Usages

S.No	Customer Name	Consumer ID	Month	Units	Bill (Rs.)	Details
1	Om Chavhan	78963215	August 2026	520.00	Rs. 2,850.00	▶ View Breakdown

Billing History - All Months & Usages

S.No	Customer Name	Consumer ID	Month	Units	Bill (Rs.)	Details			
1	Om Chavhan	78963215	August 2026	520.00	Rs. 2,850.00	▼ View Breakdown			
						Slab	Units	Rate	Amount
						First 50 units	50.00	Rs. 3.50	Rs. 175.00
						Next 100 units	100.00	Rs. 4.00	Rs. 400.00
						Next 100 units	100.00	Rs. 5.20	Rs. 520.00
						Above 250 units	270.00	Rs. 6.50	Rs. 1,755.00